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hw4_me501_Q3a.m

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```
clear all;
close all;
clc;

disp('Calculating minimum energy control sequences.')
```

```
Calculating minimum energy control sequences.
```

Setup

```
% We're simulating a discrete-time system, so all we need is a total number
% of timesteps. Call that T.
T = 3;

% Our initial condition and goal final condition:
x0 = [1; 2; -2];
xf = [0; 0; 0];

% for looping later: pick out number of states.
n = size(x0,1);

% Please put the A and B matrices from problem 2 here:
A = [0 0 0;
     0 1/2 0;
     0 0 -1/2];

B = [1; 1; 1];
```

Calculate a minimum energy sequence

```
u_Tk = soln_u_min_sequence(A, B, T, x0, xf);
% remember, the sequence of inputs is opposite in time!
% we need to apply u_kT rather than Tk.
u_kT = flip(u_Tk);
```

Simulate

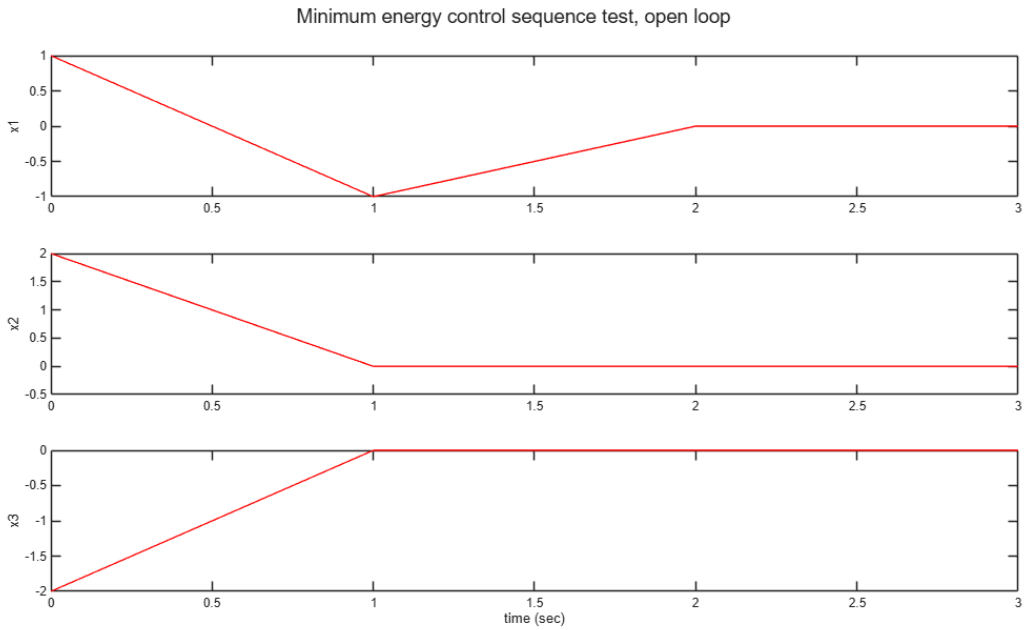
```
% insert our initial condition as the 1st element in traj
% note that traj will have T+1 columns.
x_traj = zeros(size(x0,1), T+1);
x_traj(:,1) = x0;

% Forward simulation of a discretized system is just a rollout.
% Indexing is a bit off here, we're including the initial state as the
% start of x_traj, so it's really 0:T+1 rather than 1:T.
for k=1:T
    %%%%% DYNAMICS
    x_kp1 = A*x_traj(:,k) + B*u_kT(k);
    % save the result
    x_traj(:,k+1) = x_kp1;
end
```

Plot

```
% We want to see all three states
figure;
hold on;
sgtitle('Minimum energy control sequence test, open loop')
t = [0:1:T];

for i=1:n
    subplot(n,1,i)
    plot(t, x_traj(i,:), 'r');
    % a neat trick to obtain the string "x1", "x2", etc
    ylabel(strcat("x", num2str(i)));
end
xlabel('time (sec)')
```



Verify

```
disp('x_traj final state:')
x_traj(:,end)
disp('target final state:')
xf

% Energy for this input sequence:
norm(u_Tk)^2
```

```
x_traj final state:

ans =

     -3.33066907387547e-16
     -3.88578058618805e-16
     -2.77555756156289e-16

target final state:

xf =

     0
     0
     0


ans =
```

```
1
```

Published with MATLAB® R2026a